TASK 2:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

megha@DESKTOP-JMG56TB:/mnt/c/Users/megha/OneDrive/Documents/337$ gcc -g -fno-omit-frame-pointer -O0 format_string_write2.c -o vul
format_string_write2.c: In function ‘main’:
format_string_write2.c:9:25: warning: format ‘%hn’ expects argument of type ‘short int *’, but argument 2 has type ‘int *’ [-Wformat=]
    9 |     printf("compsci469%hn", &i);
      |                        ~~^   ~~
      |                          |   |
      |                          |   int *
      |                          short int *
      |                        %n
megha@DESKTOP-JMG56TB:/mnt/c/Users/megha/OneDrive/Documents/337$ ./vul
i (before): 287454020
compsci469
i (after): 287440906
megha@DESKTOP-JMG56TB:/mnt/c/Users/megha/OneDrive/Documents/337$ gdb ./vul
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./vul...
(gdb) break main
Breakpoint 1 at 0x1175: file format_string_write2.c, line 3.
(gdb) run
Starting program: /mnt/c/Users/megha/OneDrive/Documents/337/vul

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at format_string_write2.c:3
3       int main() {
(gdb) next
4           char str[10] = "AAA";
(gdb) next
5           int x = 0x5577;
```

```
(gdb) next
6           int i = 0x11223344;
(gdb) next
8           printf("i (before): %d\n", i);
(gdb) next
i (before): 287454020
9           printf("compsci469%hn", &i);
(gdb) p &i
$1 = (int *) 0x7fffffffde94
(gdb) x/16xb &i
0x7fffffffde94: 0x44    0x33    0x22    0x11    0x77    0x55    0x00    0x00
0x7fffffffde9c: 0xff    0x7f    0x41    0x41    0x41    0x00    0x00    0x00
(gdb) next
10          printf("\ni (after): %d\n", i);
(gdb) p &i
$2 = (int *) 0x7fffffffde94
(gdb) x/16xb &i
0x7fffffffde94: 0x0a    0x00    0x22    0x11    0x77    0x55    0x00    0x00
0x7fffffffde9c: 0xff    0x7f    0x41    0x41    0x41    0x00    0x00    0x00
(gdb) exit
A debugging session is active.

        Inferior 1 [process 1283] will be killed.

Quit anyway? (y or n)
```

Str = AAA
x = 0x5577
i = 0x11223344

Memory address Before:

de94 : 44 33 22 11 (i = 0x11223344)
de98 : 77 55 00 00 (x = 0x5577)
de9c : 41 41 41 00 (Str = AAA)
deA0 : 41 41 00 00

Stack diagram Before:

| | |
|---|---|
| 00 | de9a |
| 55 | de99 |
| 77 | de98 |
| 11 | de97 |
| 22 | de96 |
| 33 | de95 |
| 44 | de94 |

Memory address After:

de94 : 0a 00 22 11
de98 : 77 55 00 00
de9c : 41 41 41 00
deA0 : 41 41 00 00

After Stack:

| | |
|---|---|
| 00 | de9a |
| 55 | de99 |
| 77 | de98 |
| 11 | de97 |
| 22 | de96 |
| 00 | de95 |
| 0a | de94 |

i = 0x1122000a